openai / **math**

Code | Pull requests | Security and quality | Insights

math

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About

No description, website, or topics provided.

Readme

Apache-2.0 license

Activity | Custom properties

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Report repository

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jrl-openai Merge pull request #1 from openai/codex/update-10-7 fd4aeeb · 4 days ago

lean	Update manuscripts and Lea...	4 days ago
preprints	Update manuscripts and Lea...	4 days ago
reasoning_traces	Initial commit	5 days ago
CONTENTS.md	Update manuscripts and Lea...	4 days ago
LICENSE	Initial commit	5 days ago
README.md	Update manuscripts and Lea...	4 days ago
history.md	Update manuscripts and Lea...	4 days ago
overview.pdf	Update manuscripts and Lea...	4 days ago
overview.tex	Update manuscripts and Lea...	4 days ago

README | Apache-2.0 license

Readme

This repository contains mathematical manuscripts and supporting proof artifacts produced by an internal OpenAI model.

As part of model development, we evaluate our models on open research problems. We expanded these evaluations after performance on our existing mathematical evaluations saturated. Some outputs build upon earlier results produced by the models.

This collection includes results at different stages of verification. Not all have accompanying Lean formalizations. We will continue to update this repository with Lean formalizations as we obtain them.

Some of the unformalized results could have issues. We will endeavor to fix any such issues quickly. We are also exploring community-hosted repositories for these materials.

Navigating the collection

The current catalogue contains 719 manuscripts organized into 372 families. A family groups related papers, which may include a principal result, companion arguments, consequences, or alternative proofs. Each family is classified by mathematical discipline.

- Start with the overview for descriptions of the families.
- Use the manuscript map to find individual papers and their supporting materials.
- The `preprints/` directory contains PDFs, source files, and manuscript-specific citation and build instructions.
- The Lean library and formalization catalogue describe the available formal proofs, their associated papers, and verification configurations. See the Comparator instructions for additional checking instructions. The repository has ~42% top-line results formalized.
- Updates to the repo are described in the history.

Reasoning summaries

We are also releasing abridged summaries of the model's reasoning, covering the following results:

Family	Subject
007	Ordinary two-point correlations of multiplicative functions
017	The irrationality exponent of π
087	Symmetric and general Mahler conjectures

Family	Subject
102	Ordinary NP-hardness at the basic semidefinite threshold
159	Quasipolynomial bounds for arithmetic progressions
197	Kaplansky's direct-finiteness conjecture in characteristic two
221	The Mézard–Parisi formula for diluted spin glasses
271	Spontaneous magnetization in the quantum Heisenberg ferromagnet
287	Isomorphism of free group factors
362	The three-dimensional relativistic Vlasov–Maxwell system

How the results were produced

The vast majority of results were obtained with the same procedure using an unreleased internal OpenAI model. On average, each result used three hours of ChatGPT Pro thinking compute with that model. Over the course of the evaluation, the model was posed approximately 4,000 problems. Aggregating the output into result families and manuscripts and requiring an appropriate level of significance led to the catalog outlined above.

Exceptions to this fixed procedure include work on a zero-free region for the Riemann zeta function and proof of the Hodge Conjecture for CM abelian varieties. Additionally, the writeup for the Re(s) > 11/12 zero-free region for the Riemann zeta function was human edited for readability.

Versions and citations

We will preserve the public release history of this collection. Corrections and revisions will be recorded as new versions, with previously released versions remaining accessible.

Releases

No releases published

Contributors 2

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Languages



Lean 91.7% TeX 8.2% Other 0.1%